

“It is important to look at technology in terms of packaging too”



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What is the latest from TI to help design engineers?

TI's WEBENCH online design centre is important for anyone designing power circuits. It basically enables the user to design circuits as models that can simulate everything online. WEBENCH design tools are software algorithms and visual interfaces that deliver complete power, lighting and sensing applications in seconds. This enables users to make value-based comparisons at system and supply chain level before a design is committed.

Since the integration of National, we have added more than a hundred TI products into the tool, and recently unveiled WEBENCH system power architect for DC-DC power supply systems. We are expanding the WEBENCH tool to design analogue front-ends and clock circuits as well.

What's the change in architecture?

It is important to look at technology not just from an IP or process node perspective but in terms of packaging too—such as the packaging of a field-effect transistor (FET), an inductor and other components inside a single device.

US-based Texas Instruments (TI) develops and commercialises analogue and embedded processing integrated circuits (ICs) and solutions. It is the third largest manufacturer of semiconductors worldwide, the largest producer of analogue semiconductors and the second largest producer of embedded processing ICs, including digital signal processors (DSPs), microcontrollers and multi-core processors.

Brian Crutcher, senior vice president and general manager, Analogue, TI Inc., recently spoke to Dilin Anand and Ashwin Gopinath of EFY about the analogue semiconductors market and TI's recent acquisition of National Semiconductor

National was very innovative in terms of packaging technology and multi-chip modules.

What are the key areas where advances are being made?

One of the key areas where we are advancing is higher voltage, such as higher-voltage FETs. We are currently working on 600-700V technologies and will reach even higher voltages in the near future. It is in a sweet spot of industrial and automotive applications, in the midst of significant growth—especially in the Indian market. So you will see us continue to innovate in high-voltage field.

Outside that, with respect to power management, we have most of the bases covered and are continuing to innovate across multiple fronts. We can get better in certain areas like hot swap and isolation, among others.

What are the challenges in hot-swap and isolation?

Customers face several design challenges in hot-swap and isolation depending on the voltage level and targeted application. Hot-swap applications require plugging of a high-powered, sensitive system board into a live (hot)